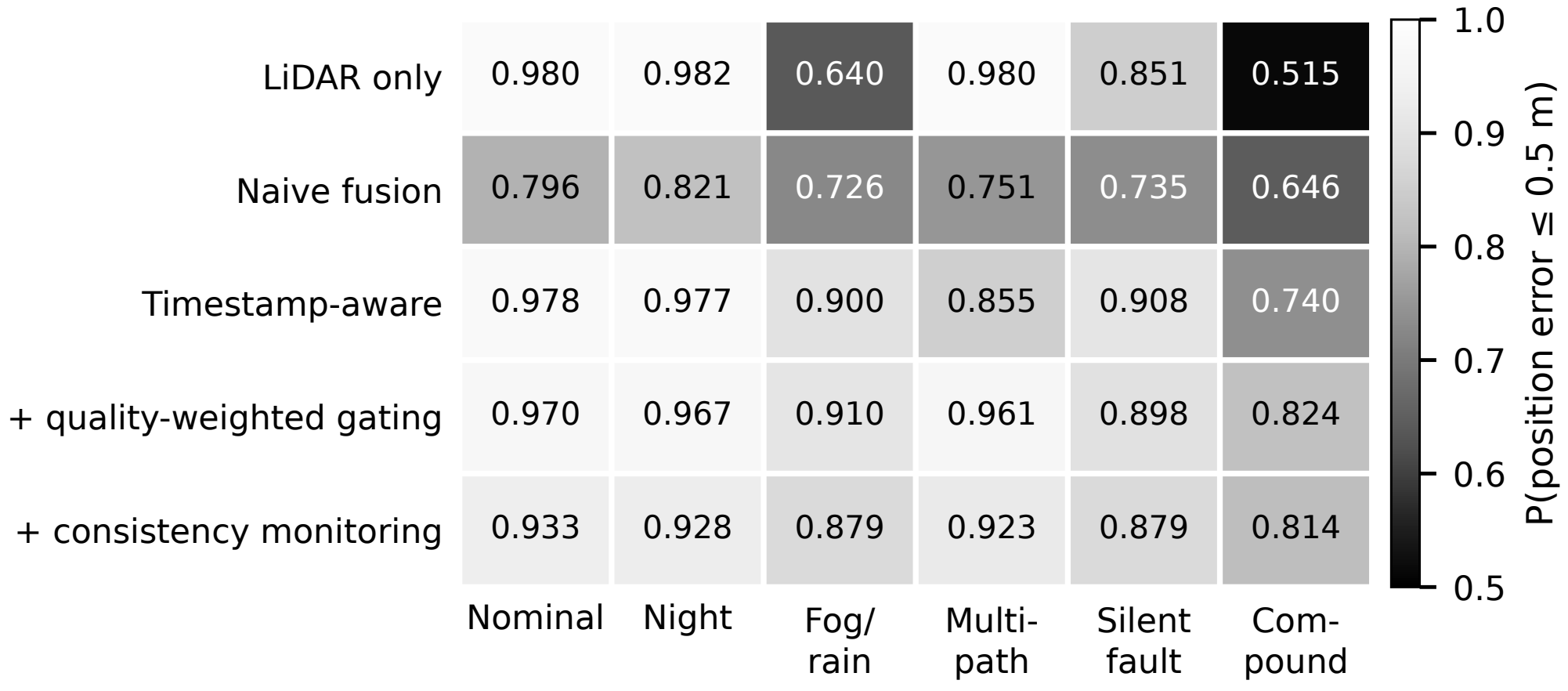
LiDAR only
Naive fusion
Timestamp-aware
+ quality-weighted gating
+ consistency monitoring
0.980
0.982
0.640
0.980
0.851
0.515
0.796
0.821
0.726
0.751
0.735
0.646
0.978
0.977
0.900
0.855
0.908
0.740
0.970
0.967
0.910
0.961
0.898
0.824
0.933
0.928
0.879
0.923
0.879
0.814
Nominal
Night
Fog/ rain
Multi- path
Silent fault
Com- pound
P(position error ≤ 0.5 m)
1.0
0.9
0.8
0.7
0.6
0.5